Siddhant Kumar Upmanyu

Senior Systems Architect

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4+ years of full ownership over backend and cloud infrastructure at an IoT smart home company. I design at the protocol level when needed — custom binary over UDP, kernel-level traffic routing, nftables packet reflection. I publish the tools I build: a Kotlin microservice framework on Maven Central, a Go assertion library, a Helm CLI in Rust, a custom Linux distro. TDD is non-negotiable across every language and every project.

EXPERIENCE

eGlu Smart Homes (WiZNSystems) — Bengaluru, India · 4+ years

Senior Systems Architect — Apr 2026 – Present

Backend & Infrastructure Engineer — Mar 2022 – Mar 2026

ARCHITECTURE & API DESIGN

- Own the API contract suite for the entire platform — 30+ specs covering device provisioning, scene/rule automation, OTA, smart lock integrations, and real-time device status. App and firmware teams build against these before a line of code is written.
- Lead technical specification for the unified backend and cloud infrastructure stack, security standards, and data contracts.

IOT PROTOCOL & REAL-TIME SYSTEMS

- Designed and implemented the custom binary protocol (UDP/TCP) between IoT hubs and cloud — packet handling, NAT hole punching, hub online/offline state machine, connection tracking, race condition handling.
- Pushed network work to the kernel when needed: nftables UDP echo server for NAT traversal (packet reflection without userspace), IP rate limiting via per-source dynamic sets at prerouting priority.
- Tuned kernel parameters for high-throughput IoT workloads: socket buffers, IP forwarding, connection tracking table size.

PLATFORM & AUTOMATION ENGINE

- Built the scene and rule automation engine: multi-hub scenes, nested fragment handling for large payloads, scene sync, IFTTT-style cloud rules, scheduling.
- Built the device provisioning system from scratch (extracted from monolith): hub setup, node validation, WiFi flow rewrite, replacement flow for failed nodes.
- Integrated Google Home, Alexa, and Yale smart locks (cloud-to-cloud) into the platform.
- Built a feature flags system for controlled rollouts.
- Introduced Consumer-Driven Contract Testing (Pact) for firmware delivery APIs — full CI pipeline: publish pact → can-i-deploy → release → mark deployed.

PERFORMANCE ENGINEERING

- ClickHouse migration: rewrote core ingestion pipeline in Go, achieved **95%+ storage reduction** (90 GB → 3.7 GB, 25 GB → 780 MB) via schema design and specialized codecs.
- JVM profiling with async-profiler and flame graphs — eliminated CPU spikes from per-packet object allocations using ThreadLocal; tuned GC for throughput.
- Zero-downtime blue-green deploys via iptables: flushed conntrack for immediate cut-over, applied rules to both PREROUTING and OUTPUT chains for internal routing.
- Implemented structured ThreadPool error handling after debugging silently swallowed exceptions in async paths (no logs, no traces — the bug that inspired the blog post).

INFRASTRUCTURE & RELIABILITY

- Migrated disk-based logging to Loki; built Prometheus + Grafana monitoring — significantly reduced MTTR.
- Architected EKS migration using Terraform; evaluated HashiCorp Nomad first, selected EKS.
- Implemented TLS client authentication with a custom CA and certificate revocation list (CRL) in Go.
- Maintained production Linux systems: kernel tuning, live Debian/Ubuntu upgrades, `tcpdump`, `strace`, `sar`, `iotop`.
- Built and maintained GitHub Actions CI/CD pipelines for automated testing, building, and deployment.

CODEBASE & QUALITY

- Migrated core monolith (Spring 4 / Java) to Spring Boot 2.7+ / Kotlin — 100% TDD throughout to prevent regressions.
- Decoupled monolith into gRPC microservices.
- Led technical recruitment: designed coding challenges, conducted interviews.

OPEN SOURCE

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stopgap Kotlin · Maven Central

Microservice framework on Helidon SE (Nima) + Project Loom. Custom compile-time DI via KSP — no runtime reflection. Three-tier testing: unit → in-process integration server → Docker E2E via Testcontainers. Gradle plugin that wires the full build pipeline. Published at `dev.sku20.stopgap:*:2.8.0`.

assertgo Go

Type-safe assertion library with generics. Fluent API, chainable `Not()`, custom matchers, zero dependencies. v1.0.0.

hrh Rust

Helm Release Helper — reads a YAML release declaration, invokes `helm upgrade --install`. Diff mode, `--atomic` rollback, chart versioning. Installable via `cargo install`.

avoid Shell

Minimal Linux distribution based on Void Linux, built for servers and recovery disks. Ships `.img.gz` and `.qcow2` images via GitHub Actions releases.

c_oop C

OOP and London-style TDD in pure C. Polymorphism via interface structs, heap-allocated objects with constructors. v5.0.0, CI.

PUBLICATIONS

Connection-Agnostic Presence Tracking for Stateless Distributed Backends Preprint · Zenodo

Redis-based architecture for tracking IoT device online/offline status across stateless distributed backends — sorted sets scored by expiry deadlines, throttled batch writes, asymmetric fault-tolerance guarantees ensuring offline transitions are never lost. Derives worst-case detection latency bounds. v1.3.0.

TECHNICAL WRITING

Writing at eglu.tech about production engineering, JVM internals, and systems design.

- Zero-Downtime Deployments with Iptables
- Java Exceptions Swallowed: The ThreadPool Trap
- ThreadLocal Optimizations and Project Loom
- A Low-Level UDP Echo Server for NAT Traversal
- What is Rate Limiting?
- Optimizing Hex Formatting
- Avoid — A Void Linux Distribution

SKILLS

LANGUAGES	Kotlin, Java, Go, Rust, Zig, C, Shell
FRAMEWORKS & PROTOCOLS	Spring Boot, Helidon SE, gRPC, Protocol Buffers
INFRASTRUCTURE	Kubernetes (EKS), Terraform, Docker, GitHub Actions, Linux (Debian/Ubuntu)
DATABASES	ClickHouse, MongoDB, PostgreSQL, Redis
OBSERVABILITY	Loki, Prometheus, Grafana, async-profiler, JMC, VisualVM, Flame Graphs
TESTING	TDD (London style), Pact (Consumer-Driven Contracts), Playwright, Testcontainers
NETWORKING	TCP/UDP socket programming, iptables, nftables, NAT traversal

READING

- Patterns of Enterprise Application Architecture
- Pattern-Oriented Software Architecture Vol. 1
- Clean Code
- Agile Software Development: Principles, Patterns, and Practices
- Test-Driven Development by Example
- Growing Object-Oriented Software, Guided by Tests
- Computer Networking: A Top-Down Approach
- Designing Data-Intensive Applications (reading)
- Elixir in Action (reading)

Readlist: Domain-Driven Design · Extreme Programming Explained